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Comparison of Hidden Markov Model and KD-Tree in GPS Data-Based Map Matching Process

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***Abstract*—Data as a source of information in decision-making can be applied across various fields, such as government policy. Naturally, government policies have a significant impact because the entire society can feel their effects. Therefore, careful planning and structured implementation are required. Data can serve as one of the parameters for accurate planning. Policies can cover a wide range of scopes, including public mobility. An example of such a policy is the emergency health facilities built in the Java and Bali regions during the COVID-19 pandemic. Various types of data can be used in designing these policies, one of which is mobile positioning data (MPD). This data is advantageous because it can be collected quickly. However, MPD requires a complex process due to noise that can affect the information produced. One method to address this noise is map matching. The author is interested in comparing the accuracy levels of map matching using the Hidden Markov Model (HMM) algorithm and the KD-Tree algorithm. In this study, both algorithms process different categories of GPS data. Additionally, there is manipulation of the GPS data used, specifically random reduction. The parameter used to compare these algorithms is the similarity score obtained from the distance values in the Dynamic Time Warping (DTW) method. The test results show that the HMM algorithm has an accuracy level of about 80% for all tested data categories, while the KD-Tree algorithm has an accuracy level of about 55% for all tested data categories. It can be concluded that the HMM algorithm is superior in performing map matching for all data categories compared to the KD-Tree algorithm. Furthermore, it was found that a random reduction of GPS data can affect the resulting accuracy levels.**

***Index Terms*—GPS Data, Map Matching, Hidden Markov Model, KD-Tree, Dynamic Time Warping**

I. Introduction

In the current digital era, the expression ”data is the new gold” is frequently used. Data might reveal patterns that are not always obvious right away. With the knowledge gained from these patterns, data may start to play a key role in decision-making procedures. Government policy is one of the many domains where data can be used as an information source for decision-making. Human mobility data is one kind of data that can be used in the context of government policy. This data shows how people travel from one location to another. Budiono et al. indicate that the government's Covid-19 pandemic policy is one example of how human mobility data was used [1]. One important factor in the decline of Covid-19 cases was the establishment of emergency medical centers in the Java and Bali regions. This happened as a result of emergency health facilities being planned and operated using knowledge derived from available data.

Data about human mobility can be obtained in several ways. One approach is to conduct surveys by directly asking respondents. However, because gathering the data takes a significant amount of time, this strategy is seen as relatively ineffective. Furthermore, biases and mistakes can occur with this strategy [2]. With the development of modern technology, mobile positioning data (MPD) offers an additional method of acquiring mobility data. Positional information collected from a user's mobile device is called MPD. There are two varieties of this positional data: active and passive. Passive MPD is a technique for automatically gathering positioning data depending on the user's cellular operator operations, such as sending messages and making calls, according to research by Park et al. [3].

To summarize, the MPD methodology is one method for obtaining population mobility data. However, both active and passive MPD approaches are subject to noise, which can have an impact on the quality of the information obtained from the data. To improve the accuracy of the generated information, more research is needed to reduce the impact of noise on GPS data. Map matching is a technique for processing this data, which often involves altering GPS data that falls outside of roadways to align it within them, making movement pathways more readable. Map matching employs a variety of techniques, each of which is designed to handle unique concerns with GPS data. The contribution of this paper is to compare how well the KD-Tree and Hidden Markov Model (HMM) algorithms perform map matching on GPS data.

II. Related Work

Yuan et al. classify map matching techniques into five categories: path distance, point distance, probability-based, model-based, and learning-based. Each category has a different approach to dealing with GPS noise and produce more accurate results [4].

Using Hidden Markov Models (HMM), Newson et al. utilize active GPS MPD data from an 80-kilometer trip in Washington, USA, for their map matching experiments [5].

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Using this technique, Matteo et al. [6] developed PyTrack, a Python library that uses Newson's HMM algorithm to match maps.

Zhu et al. point out that global map matching methods are computationally less efficient but can achieve higher accuracy than incremental methods [7]. They develop Longest Common Subsequence (LCS) algorithm using data from 100 trips.

Lu et al. focus on the problem of low GPS sampling rates, or data points separated by relatively large intervals [8] . Their work used the ST matching technique to consider factors such as the likelihood of a straight route rather than a winding one and speed limits on roads, thereby improving the accuracy of map matching.

Rakhman et al. use the KD-Tree and A* algorithms for their map matching procedure [9]. However, the lack of ground truth or actual routes in the test data made it difficult to evaluate the accuracy of the algorithms.

In addition to specific algorithms, machine learning techniques for map matching were also investigated. Sinsopvalagul et al. use machine learning techniques to match maps using multiple classification models [10].

III. Methodology

A. Data

The author examines several datasets that need to be processed to evaluate how effectively the algorithms HMM and KD-Tree perform during the map matching process. Latitude and longitude are typically the features of all the GPS data categories that the map matching algorithm needs to verify.

1) Author's Travel Data: Travel on the roadways of Depok City, West Java, Indonesia, is simulated by the author in this category. GPS Logger is an application that tracks the user's position and path, according to information provided by the program's developer, BasicAirData [11]. Numerous data points are recorded during the recording process, including longitude and latitude. There are parameters in the GPS Logger that can impact the outcome of the recording. Two parameters are changed in this study: the distance filter and the interval filter. The interval filter is a setting that determines the recording of GPS data based on a specified time interval. In this study, the interval filter is set to its maximum, which is 1 second. The distance filter is a setting that determines the recording of GPS data based on a specified distance. In this study, the distance filter is set to its maximum, which is 0 m.

2) Research Data from Newson et al. [5]: Newson et al. [5] used travel data from Seattle, Washington, USA, for their analysis. The RoyalTek RBT-2300 GPS logger hardware was used to record the journey routes, according to literature found on the Microsoft website [12]. Three categories of data are employed in the study by Newson et al. [5]: GPS data from the test trip, data from the road network, and ground truth path.

3) Synthetic Data: The author used a website called GPS Visualizer to duplicate the journey route that was captured by the GPS Logger application to create ground truth data. Every sort of input is displayed on a map with travel routes formed by connecting particular spots on the road. The coordinates of the locations that make up a route can be exported after they have been defined.

4) Road Network Data: Road network data is a collection of coordinates that represent a network of roads. Using this information, a graph is created that is used in the GPS data matching process. The road network data was gathered from the Hot Export Tool website, which offers open services for exporting OpenStreetMap data. There are various types of data, including information about bridges, buildings, and infrastructure. However, only data from road segments was used in this study.

B. Hidden Markov Model

The Hidden Markov Model (HMM) is a probabilistic method for modeling data sequences. HMM fundamental idea is the Markov chain, which characterizes a system's transitions. This system is made up of states that shift from one to another. The transition process depends only on the prior state and does not consider other states. This notion is commonly referred to as the Markov property [13]. The map matching process falls within the Markov chain since it meets the Markov property. The determination of GPS data transitions associated with a road section is exclusively based on previous GPS data. The Hidden Markov Model is classified as a state-transition model with map matching. This is because HMM employs topology in the map matching process, specifically a directed graph.

The workflow for the HMM method is shown in Figure 1. The data from Newson et al. [5] and the author's trip data are processed using this approach. The PyTrack package is used to implement the HMM technique. Map matching is a feature of the Python library PyTrack, according to research by Tortora et al. [6].

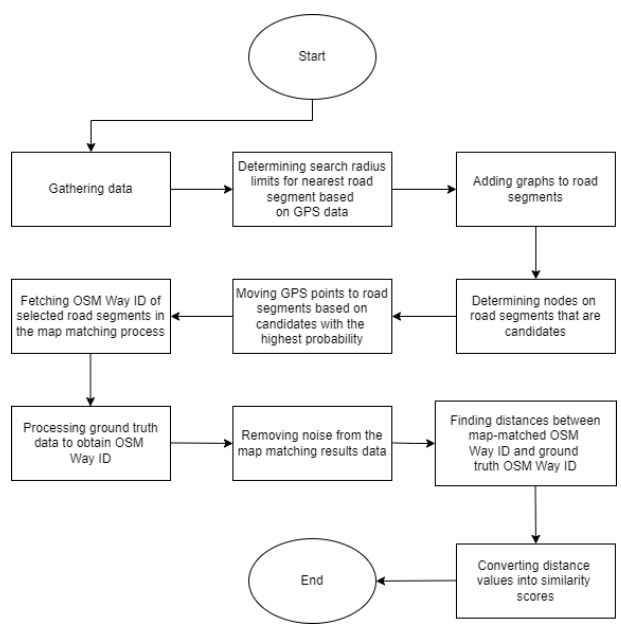

Fig. 1. Hidden Markov Model Method Flowchart

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First, GPS data is gathered. Two types of GPS data are gathered for the Hidden Markov Model method: the author's journey data from the GPS Logger application and the data from Newson et al.'s study [5].

Second, based on a predetermined radius, GPS data is matched with the closest road segments. More road segments will be matched with the GPS data the higher the radius value chosen, which might lead to a significant rise in processing needs. Furthermore, the likelihood of a match between the GPS data and a road segment decreases with increasing distance between the road segment and the GPS location being analyzed.

Third, the previously established region is expanded to include a graph. It takes a map with a graph on every road segment to carry out the map matching procedure. The GPS data will be compared with several nodes and edges that represent each road segment graph. Using a method in the PyTrack library, the predetermined region is downloaded as a map based on the boundaries that have already been established. Subsequently, as Figure 2 illustrates, a graph made up of nodes and edges is added to the map.

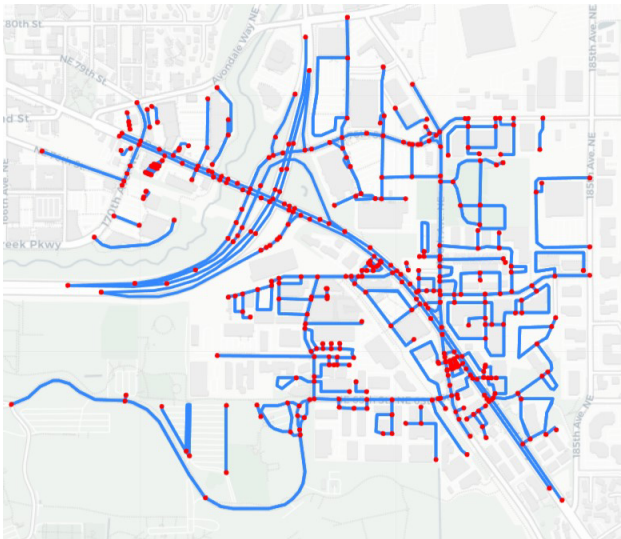

Fig. 2. Map Display with Graph Using HMM Method

Fourth, determining candidate nodes. The graph of each road segment is made up of several nodes that are going to be compared with GPS information. To find the nearest road segment, the nodes that are closest to the GPS data are chosen as references. The author uses PyTrack library methods in this procedure. The result of using the function is shown in Figure 3.

Fifth, moving GPS data to road segments based on candidates with highest probability. The chosen candidates will go through additional processing in order to identify nodes that will serve as references for comparing GPS coordinates and closest road segments. The probability of each candidate combination varies. The result of map matching will be the road route with the highest likelihood. The resultant road route's visualization is displayed in Figure 4.

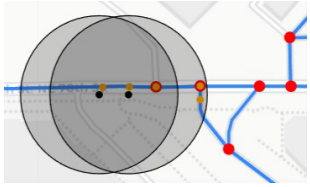

Fig. 3. Display of Candidate Search Results in HMM Method

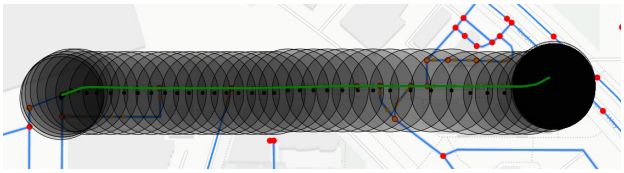

Fig. 4. Display of Road Routes from HMM Map Matching Process

Sixth, getting the OSM Way ID from the road segments that map matching produces. In the map matching procedure, the chosen road route is made up of many nodes with chosen road segments. This data consists of representations, node IDs, and OSM Way IDs—identifiers provided by OpenStreetMap to indicate physical locations like highways.

Seventh, processing ground truth data to obtain OSM Way ID. In this work, two ground truth datasets are processed. The first dataset is the ground truth data for the author's travel data, while the second dataset is used in the study by Newson et al. [5]. GPS Visualizer is used to create the ground truth travel data for the author. Using the Nominatim API via the GeoPy library, the two ground truth datasets will go through the same phases. OpenStreetMap Nominatim has the ability to convert coordinate points into addresses for a location, as stated in Serere's diary [14].

Eighth, removing noise from the map-matching results. Noise was found in the OSM Way ID data obtained from map matching, which could have an impact on accuracy levels.

Ninth, calculating distances value. These numbers are used as parameters to evaluate how close the patterns are between the ground truth data and the OSM Way ID data that are obtained using map matching. The DTAIDDistances library is used in the distance computation process. In order to derive similarity scores, distance values are further analyzed.

Tenth, converting distance values to similarity scores. To determine similarity scores, additional processing is done on the distance values that were acquired in the preceding phase. The proportion of pattern similarity between the ground truth data and the OSM Way ID data that results from map matching is represented by the similarity score.

C. KD-Tree with A*

A KD-Tree, as defined by Robert [15], is a data structure for organizing multidimensional data. A map's two-dimensional coordinates, notably longitude and latitude, are an example of multidimensional data. KD-Trees can handle not only two-dimensional data but also three-dimensional and higher-dimensional data. A variety of things must be addressed while

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organizing data using the KD-Tree technique. Examining these nodes' levels is crucial when comparing a new data point to ones that are already in the KD-Tree structure. When comparing two-dimensional data, even levels compare the values of the y-coordinate, and odd levels compare the values of the x-coordinate. After the comparison, data points with lower values than the previous node are placed on the left side, while those with higher values are placed on the right.

The A* algorithm, as explained by Zhang et al. [16], determines the shortest route from start to end using heuristic search, which can use parameters like Manhattan distance, Euclidean distance, and Chebyshev distance. Traffic density, which is obtained from each node in the map network, is the heuristic parameter used by Rakhman et al. [9] The optimal route is found using the A* algorithm once a road section and GPS data are matched. Using coordinate data from map-matched results, computing distance values, translating them into similarity scores, processing coordinate data to produce OSM Way IDs, and removing noise from the map-matched data are the next steps, which are comparable to the HMM technique.

Figure 5 shows the flowchart for the KD-Tree and A* approach. The author's trip data and the research data of Newson et al. [5] are processed using this flow. The HMM approach and the first step are similar. Both the travel information logged by the GPS Logger application for the author's trips and the information utilized in the Newson et al. study [5] comprise the same set of data. The road network data for the Seattle and Depok City regions must be acquired from the Hot Export Tool website in addition to GPS data in order for the KD-Tree and A* techniques to function.

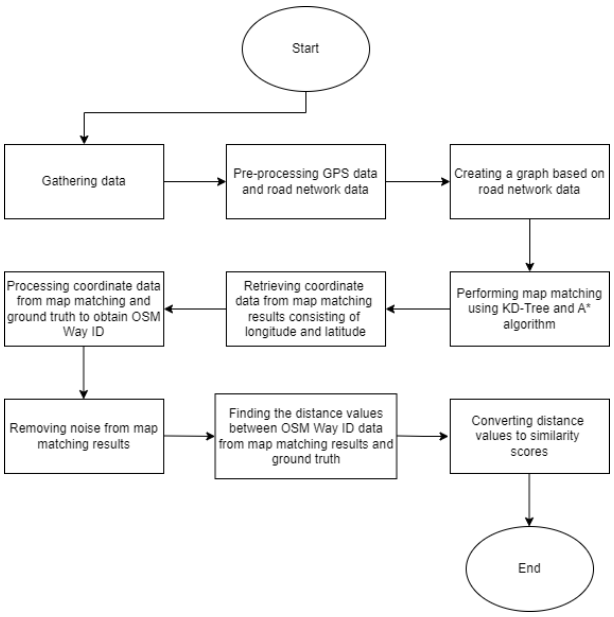

Fig. 5. Flowchart of the KD-Tree and A* Method

Second, drawing a graph using the data from the road network. Rakhman et al. [9] state that the orientation of the road segments is taken into consideration when building the graph that serves as the foundation for the map matching procedure. Road segments will be chosen so as to match GPS data by taking into account their sequential direction. Using the NetworkX package, a road network graph is produced.

Thirdly, employing KD-Tree and A* to match maps. The map matching procedure is applied to GPS data using the road network graph that was generated in the preceding stage. The KD-Tree technique converts the road network graph into nodes. The scipy library is used in the tree construction process of this technique. The closest road segment is then identified by nodes by processing GPS data with the resulting tree. The next stage is to use the A* algorithm to determine the best travel route after GPS data has been matched with a particular road segment. Subsequently, the chosen path is displayed. The map matching results utilizing the KD-Tree and A* approaches are illustrated in Figure 3.18.

Fig. 6. Map Matching using *KD-Tree* dan A*

The following steps are similar to the ones in the HMM method: taking coordinate data from the map-matched results; calculating distance values; converting distance values into similarity scores; processing coordinate data from map-matched and ground truth data to obtain OSM Way IDs; and eliminating noise from the map-matched data.

D. Evaluation Metrics

The algorithms' performance is measured by computing the similarity score using the Dynamic Time Warping (DTW) approach. This method is particularly useful for comparing the results of map matching to ground truth data. The similarity score is calculated using the DTW distance and represents the degree of similarity as a percentage, with larger values reflecting greater likeness.

1) Dynamic Time Warping Distance: Dynamic Time Warping (DTW) is a technique for determining the similarity of two time series data sets. Because DTW is unaffected by time shifts, it can recognize similar patterns in time series data

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even when the sequence timings change. The results of map matching are one of several types of time series data. As a result, the DTW technique is effective when comparing map matching results to ground truth data. Equation 1 measures the similarity or difference between two data sets. When the distance value is almost zero, it indicates that the patterns in the compared data sets are very similar.

$$DTW_{(x,x')} = |A_i - B_j| + \min(D[i-1,j-1], D[i-1,j], D[i,j-1]) \quad (1)$$

- x: The first dataset to be compared
- x': The second dataset to be compared
- i: The sequence of data in the first dataset
- j: The sequence of data in the second dataset
- D: The distance value

2) Similarity Score Dynamic Time Warping: When assessing pattern similarity in data sets with large values, the distance measure is limited. This problem stems from the fact that the distance equation uses the difference between two data points, which, in the event that the difference is significant, produces a huge distance number. As such, it is difficult to obtain a distance value that is almost zero. When comparing OSM Way IDs between the ground truth data and map matching results, this situation arises. The distance value can be transformed into a form known as the similarity score, which is independent of the size of the data being compared, to solve the issue of huge data sets. Li et al.'s research [17] states that equation 2 is used to determine the similarity score. This equation yields a percentage with a range of 0 to 100. The distance value and maximum distance value in this study are computed by the author using the DTAIDDistances library [18]. The path that is chosen determines the difference between the distance value and the maximum distance value. The distance number for any dataset takes into account the smallest distances between any two data points. However each dataset takes the maximum value into account in order to get the maximum distance value.

$$\text{Sim} = 1 - \frac{d}{D} \quad (2)$$

- d: The distance value
- D: The maximum distance value

IV. Results

The similarity scores of the map matching outcomes obtained from the KD-Tree and Hidden Markov Model (HMM) approaches can be seen in Table I. The table shows that the HMM approach performs better in map matching outcomes than the KD-Tree method in the research data by Newson et al. [5]. When compared to KD-Tree, HMM displays significantly lower distances, indicating that the patterns in its map matching findings are closer to the actual data than those of KD-Tree.

The HMM method's distances are less than the KD-Tree method's in the author's travel data. This shows that, when compared to KD-Tree, map matching results utilizing HMM align more closely with the ground truth. Furthermore, the HMM approach has a far greater similarity score than the KD-Tree method.

TABLE I
Accuracy Rate of Map Matching Results

Percent Data	Method	Data Source	Distances	Similarity Score
100%	HMM	Newson *et al.* [5]	8.68B	87.077%
		Author's data	830.98M	87.189%
	KD-Tree	Newson *et al.* [5]	73.53B	64.952%
		Author's data	3.15B	55.420%
75%	HMM	Newson *et al.* [5]	8.03B	86.172%
		Author's data	830.70M	85.226%
	KD-Tree	Newson *et al.* [5]	63.09B	63.995%
		Author's data	3.21B	53.937%
50%	HMM	Newson *et al.* [5]	7.78B	83.780%
		Author's data	968.53M	79.460%
	KD-Tree	Newson *et al.* [5]	52.85B	63.797%
		Author's data	3.48B	51.158%

Therefore, the HMM technique is considered to be better at map matching than the KD-Tree method based on both the research data from Newson et al. [5] and the author's travel data. The HMM method's decreased distance values and greater similarity scores support this finding.

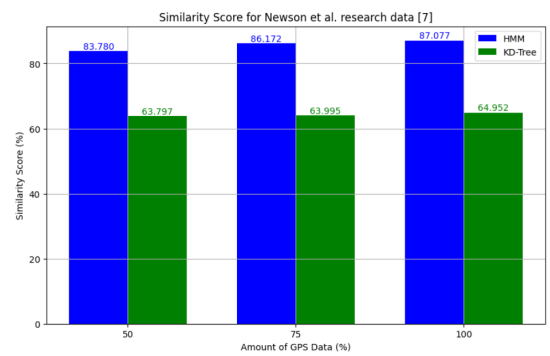

Fig. 7. Comparison of Similarity score between HMM and KD-Tree methods in various data percentages (50%, 75%, 100%). Higher values indicate better performance.

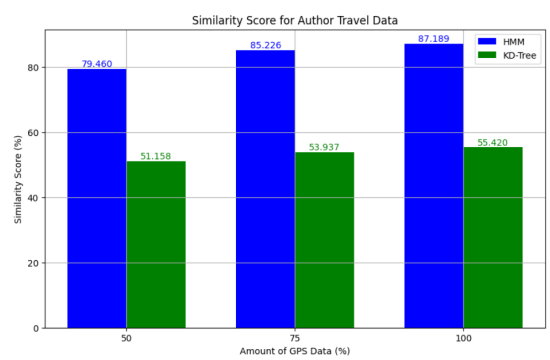

Fig. 8. Comparison of Similarity score between HMM and KD-Tree methods in various data percentages (50%, 75%, 100%). Higher values indicate better performance.

GPS data was randomly deleted in order to see how the

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quantity of data affected the degree of map matching accuracy. There were 25% and 50% of deleted data points, which translated into processed GPS data amounts of 75% and 50%. Figure 7 and 8 show that there is a positive link between the decrease in GPS data and the accuracy level of the map matching findings using the HMM and KD-Tree methods. The accuracy level decreases with the amount of GPS data handled. Furthermore, for all categories of data amounts, the author's trip data and the study data of Newson et al. [5] show a greater accuracy level when using the HMM approach. For each approach, the distinction in accuracy levels between the data quantity categories is not highly significant.

V. Conclusion

There are a number of ways to approach process map matching, including KD-Tree algorithms and the Hidden Markov Model (HMM). Both algorithms handled two types of data in the author's study: the author's trip data captured by a GPS Logger application and the research data of Newson et al. [5]. The HMM and KD-Tree algorithms produce different implementation phases, but they both result in the same output—the journey path. For every data category, these travel itineraries are then contrasted with ground truth data. In the comparison procedure, ground truth and OSM Way ID data from the map matching results are used to determine distances using Dynamic Time Warping (DTW), which indicates how similar the data patterns are between the two. On the other hand, the distances are highly impacted by the presence of very big OSM Way ID values, which results in inconsistent values. Therefore, further distance processing is required to get more consistent measures. These distances are converted into similarity scores in the author's study.

The map matching capabilities of the HMM and KD-Tree algorithms have been demonstrated using the research data of Newson et al. [5] as well as for the author's GPS-logged journey data. The lesson learnt are as follows.

1) The Hidden Markov Model approach outperforms the KD-Tree technique in map matching for two datasets. In Newson's dataset, a set with significant amount of GPS data, the HMM algorithm has a significant difference of about 20%. In authors' dataset, which contains less GPS data points, The HMM approach performs better than the KD-Tree technique by a greater margin, about 30%. Even with random data reduction applied (reduced to 75% and 50%), the HMM algorithm consistently outperforms the KD-Tree algorithm by a comparable margin.
2) The accuracy level that results is greatly affected by the reduction in processed GPS data. According to the study's findings, accuracy levels drop as GPS data volume falls.

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